

PRODUCTION ARCHITECTURE — REFERENCE

Workout Tracker deployment

How a commit becomes a running server, and how the server stays running: the build/deploy pipeline and the runtime it lands on, in one place for when it's time to build this out.

Hetzner CX22

Docker Compose

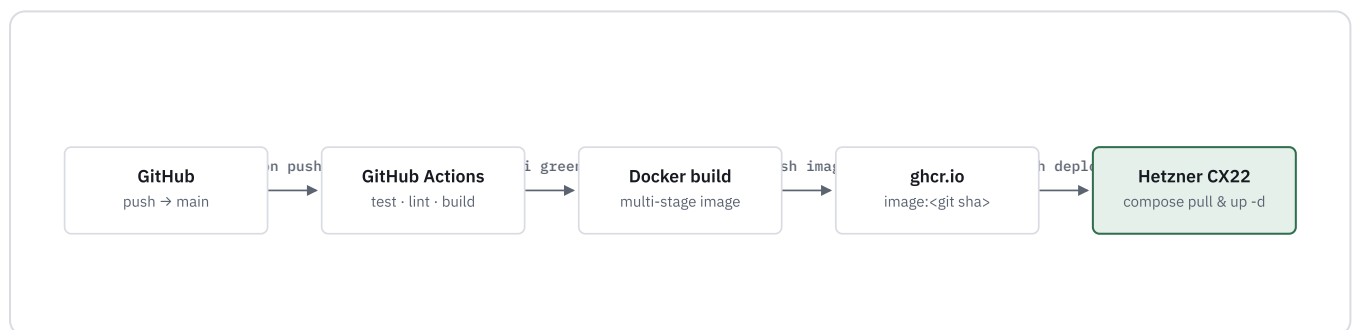
GitHub Actions

Caddy

SQLite (better-sqlite3)

01 Build & deploy pipeline

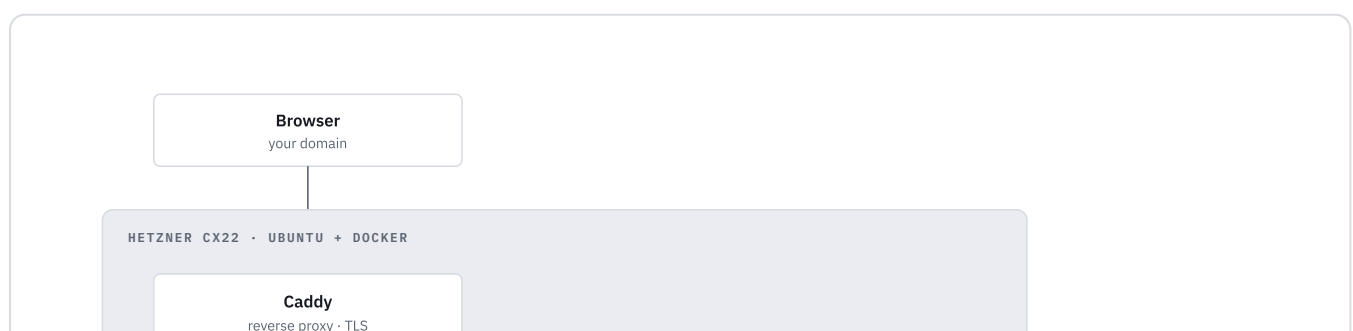
Runs in GitHub Actions. Nothing reaches the server that hasn't passed tests and been built into an image first.

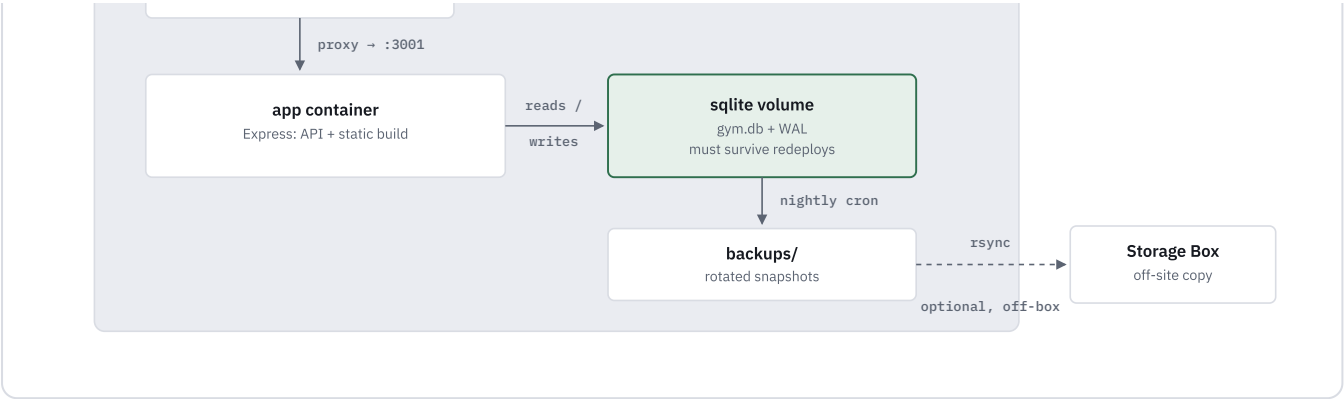


Local machines never build or ship the image — CI does, tagged by commit SHA, so what's running in production is provably what passed tests.

02 Production runtime

One app container does double duty — it serves the built React static files and the /api routes from the same process, so there's no separate client container or CORS to configure.





The volume is the one piece of state that isn't reproducible from git — everything else in the outer box can be destroyed and recreated by a redeploy.

03 Reference

! Set `DB_PATH` before the first deploy. `server/db.js` defaults to a path next to the source code — fine for dev, but inside a container that path dies with the container. Point it at the mounted volume instead, e.g. `DB_PATH=/data/gym.db`, and mount the volume at `/data`.

Give production its own `JWT_SECRET` — don't carry over the dev value from `.env`.

Environment variables

NAME	SET WHERE	NOTES
PORT	.env on host	Internal only — Caddy is what's actually exposed.
DB_PATH	.env on host	Must point inside the mounted volume, e.g. /data/gym.db.
JWT_SECRET	.env on host	Production-only value, injected at container runtime — never baked into the image.
JWT_EXPIRES_IN	.env on host	Session length, unchanged from dev unless you want otherwise.
CLIENT_URL	unset in prod	Only matters for CORS across origins — not needed once the app serves its own static build.

Volumes & ports

WHAT	WHERE	NOTES
gym-data	/data in app container	Holds gym.db, -wal, -shm. The thing that must not be lost.

caddy-data	/data in Caddy container	TLS certificates — losing it just means Let's Encrypt re-issues.
80 / 443	host → Caddy	Only ports open to the internet.
3001	Caddy → app (internal)	Not published on the host — reachable only inside the compose network.